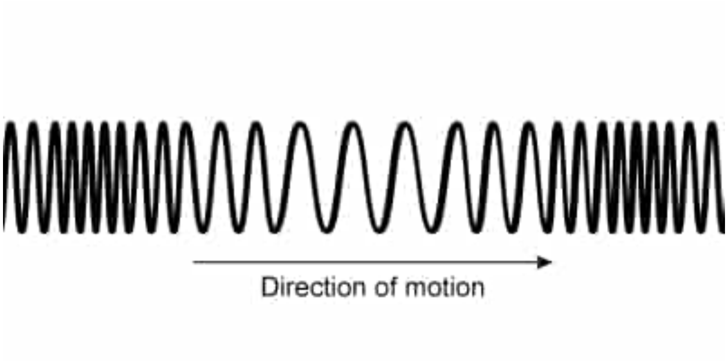
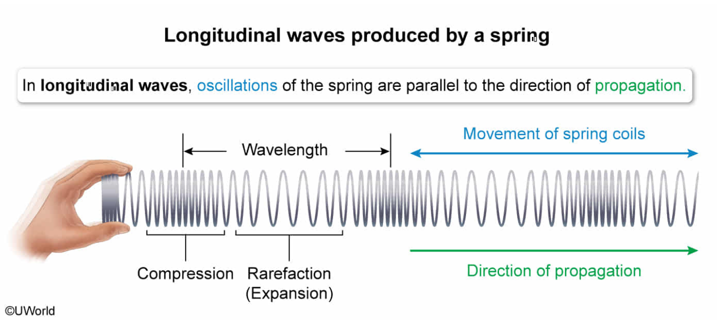


The figure above shows a mechanical wave traveling to the right on a spring. Can the wave be classified as a longitudinal wave?

- A. Yes, because the wave's direction of motion is to the right. [5%]
- ✔ B. Yes, because points on the spring oscillate along the wave's direction of motion. [64%]
- C. No, because the wave is transverse; points on the spring oscillate perpendicular to the wave's direction of motion. [28%]
- D. No, because the wave shown is a standing wave. [1%]

Correct
64% answered correctly

Explanation:



A mechanical wave is a disturbance that moves through a medium (eg, air, water). Mechanical waves can propagate through a medium in two different wave forms: transverse and longitudinal.

Transverse waves displace the components of the medium **perpendicular** to the direction of the wave's propagation. An example of a transverse wave is a water wave in the ocean, where the water

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rises and falls vertically relative to the ocean surface, but the wave travels horizontally parallel to the surface.

In contrast, **longitudinal waves** displace the medium in a direction **parallel** to the direction of the wave's propagation. For example, sound waves create disturbances in the air, causing air molecules to oscillate back and forth parallel to the direction the sound wave travels through the air.

In this question, the spring coils in the figure (the medium) oscillate between regions of compression (shown as darker areas) and regions where they are stretched out. The direction of these oscillations is horizontal, parallel to the direction of the wave's motion. Therefore, the wave produced by the oscillating spring is a longitudinal wave.

(Choice A) A wave's direction of propagation is not enough to determine whether it is longitudinal or transverse. Information about the direction of wave oscillation is also needed.

(Choice C) The points on the spring oscillate in a direction parallel to the direction of the wave's propagation, not perpendicular.

(Choice D) Standing waves are created when waves reflect onto themselves and create interference patterns that make it appear as if the wave is not propagating.

Educational objective:
Longitudinal waves displace a medium parallel to the direction of the wave's propagation, but transverse waves displace a medium perpendicular to the direction of the wave's propagation.

Time spent:0 sec
Subject:
Physics

QID:403955
System:
Light and Sound

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